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# Class discrepancy-guided sub-band filter-based common spatial pattern for motor imagery classification

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ABSTRACT

*Background:* Motor imagery classification, an important branch of brain-computer interface (BCI), recognizes the intention of subjects to control external auxiliary equipment. Therefore, EEG-based motor imagery classification has received increasing attention in the fields of neuroscience. The common spatial pattern (CSP) algorithm has recently achieved great success in motor imagery classification. However, varying discriminative frequency bands and few-channel EEG limit the performance of CSP.
*New method:* A class discrepancy-guided sub-band filter-based CSP (CDFCSP) algorithm is proposed to automatically recognize and augment the discriminative frequency bands for CSP algorithms. Specifically, a priori knowledge and templates obtained from the training set were applied as the design guidelines of the class discrepancy-guided sub-band filter (CDF). Second, a filter bank CSP was used to extract features from EEG traces filtered by the CDF. Finally, the CSP features of multiple frequency bands were leveraged to train linear support vector machine classifier and generate prediction.
*Results:* BCI competition IV datasets 2a and 2b, which include EEGs from 18 subjects, were used to validate the performance improvement provided by the CDF. Student's *t*-tests of the CDFCSP versus the filter bank CSP without the CDF showed that the performance improvement was significant (i.e., *p*-values of 0.040 and 0.032 for the ratio and normalization mode CDFCSP, respectively).
*Comparison with existing method(s):* The experiments show that the proposed CDFCSP improves the CSP algorithm and outperforms the other state-of-the-art algorithms evaluated in this paper.
*Conclusions:* The increased performance of the proposed CDFCSP algorithm can promote the application of BCI systems.

## 1. Introduction

In recent years, due to the promotion of brain science, cognitive science, electronic engineering technology and computer information technology, research on brain-computer interfaces (BCIs) has developed rapidly and received increasing attention from researchers. BCIs provide a new method for the brain to communicate directly with computers or other external electronic devices, thus allowing users to communicate with the outside world by merely relying on their brain consciousness without performing actual movement. Such communication is beneficial for patients who have lost their ability to control parts of their body (Wolpaw et al., 2002). Motor imagery classification is an important branch of BCI research, and many BCI systems have been built based on motor imagery analysis.

Electroencephalography (EEG) provides information on the functional state of the brain. Because of the high temporal resolution (Khan et al., 2014), low material and device costs, and lessening environmental restrictions, EEG signals are widely used in BCI systems. Moreover, many fast and reliable EEG-based signal-processing methods have been proposed to preprocess data and extract features (Lu and Yin, 2015). The main problem associated with EEG-based BCI systems is that the EEG signal is weak and susceptible to noise and artefacts, such as 50 Hz power frequency interference, circuit noise, and physiological signals, such as electrooculography (EOG), electromyography (EMG), and electrocardiography (ECG). In addition, motor imagery-based BCI systems are implemented to detect event-related desynchronization

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(ERD) and event-related synchronization (ERS) (Pfurtscheller and Neuper, 1994) over the motor cortex area; however, the ERD/ERS signals between different subjects vary in frequency band stimulation, location in the brain, and signal pattern (Guger et al., 2000). Thus, designing a spatial filter for the subject is particularly important for the realization of BCI systems. A good spatial filter can provide a higher signal-to-noise ratio (SNR) so that it may achieve a better classification rate and provide a higher information transform rate (ITR).

The common spatial pattern (CSP) algorithm has achieved great success in BCI system applications (Park et al., 2017). CSP is a spatial filter designed to maximize the variance in one class and minimize the variance in another class simultaneously (Lemm et al., 2005). Therefore, CSP-based approaches are fairly effective in two-class motor imagery classification-based BCIs. Many novel algorithms based on a modified CSP have been proposed (Lu et al., 2009; Kang and Choi, 2014; Su et al., 2015). Li et al. integrated motor relevant information to suppress the influence of irrelevant information and noise to improve the CSP performance (Li et al., 2017). If the CSP is applied in a small-sample setting situation, unreasonable covariance matrix estimations lead to limited performance; thus, the regularized CSP (R-CSP) algorithm was proposed to address this problem (Lu et al., 2010). Kumar presented a single band CSP framework for MI-BCI that utilizes the concept of tangent space mapping (TSM) in the manifold of covariance matrices (Kumar et al., 2017a). To reduce the variance and bias of the estimated covariance matrix, the covariance matrixes of the target subject and other subjects were combined using regularization parameters. Experiments showed that the R-CSP algorithm significantly outperformed other methods in a small-sample setting. Because the frequency range reflecting ERD/ERS varies from subject to subject, the most discriminative frequency bands for CSP varies as well. Therefore, frequency bands or feature selection for each subject plays a vital role in distinguishing these patterns (Ang et al., 2012; Luo et al., 2016; Thomas et al., 2009). The sub-band CSP (SBCSP) and filter bank CSP (FBCSP) have been implemented so that the CSP could be employed in different frequency bands, and then the features from multiple frequency bands are combined to obtain the final classifier. SBCSP uses multiple sub-band filters to preprocess the EEG before extracting CSP features (Novi et al., 2007) and fully exploits the information in different frequency bands, and it has obtained very promising experimental performance. Similarly, the FBCSP (Ang et al., 2012) uses 9 sub-band bandpass filters covering the 4 to 40 Hz frequency range to process signals. After filtering, the CSP algorithm is utilized to extract features. And then, mutual information-based best individual feature (MIBIF) and mutual information-based rough set reduction (MIRSR) are applied to select the most discriminative features. Finally, the selected features for different subjects are fed into the naïve Bayesian Parzen window (NBPW) classifier. Because of the outstanding performance, FBCSP scored first place with the BCI competition IV dataset 2a and 2b (Tangermann et al., 2012). Park et al. incorporated the ideas of FBCSP and R-CSP to propose a filter bank regularized CSP, and the experiment showed an obvious improvement in performance in a small-sample setting situation (Park et al., 2017).

Many researchers have proposed discriminative frequency bands selection based methods. The discriminative common spatial pattern (DCSP) method applied the fisher ratio to select the frequency bands with better discriminating power and the discriminative frequency bands of each subject were used for further feature extraction (Thomas et al., 2009). An improved discriminative filter bank selection approach was proposed to select the discriminative filter bank based on mutual information (Kumar et al., 2017b). The temporal filter parameter optimization with CSP approach (TFPO-CSP) tuned the parameters of a butterworth filter to improve the performance of CSP feature (Kumar and Sharma, 2018). The fisher WPD-CSP algorithm discriminated the helpful sub-bands for each subject by fisher distance, and combined the wavelet packet decomposition (WPD) and CSP to extract effective features (Yang et al., 2016). Zhang introduced a new method that implements sparse Bayesian learning of frequency bands for feature selection and classification (Zhang et al., 2017). Miao proposed a novel spatial-frequency-temporal optimized feature sparse representation-based classification method, in which significant CSP features on frequency-temporal domains were selected automatically (Miao et al., 2017).

Traditional approaches select the most discriminative frequency bands for specific subject, but these selection procedures would not be able to strengthen the discriminant ability of features. How to enhance the discriminant ability of the CSP features based on the discriminative frequency bands is a key factor for classification performance improvement.

On the other hand, the number of channels utilized in EEG collection greatly affects the performance of the CSP algorithm. For CSP-based algorithms, an EEG with more channels can easily obtain a good performance. However, in certain situations, an EEG with only a few channels is employed in motor imagery classification. But CSP feature based on few-channel EEG usually leads to poor performance in classification. Determining a method of reasonably extending the EEG channel number has become a key problem.

To overcome the weaknesses mentioned above, class discrepancy-guided sub-band filter (CDF) is proposed in this paper. CDF enhances the discriminant ability of the CSP features by augmenting the signal in discriminative frequency bands. To the best of the authors' knowledge, previous work has not been performed to augment the discriminative frequency bands for CSP based approach efficiently. Moreover, the CDF extends the channel number of the EEG reasonably. Inspired by the target guided sub-band filter for acoustic event detection (Feng et al., 2015), the CDF takes advantage of the a priori knowledge of the trials in each class, and the differences in the classes' templates are used as a guide for the sub-band filter. First, the energy spectrum extracted by wavelet packet decomposition (WPD) is averaged to obtain the class template, and the template differences between classes are applied to guide the design of the CDF. Second, the CDF is leveraged to filter the segmented EEG. Then, the FBCSP is employed to extract features from the filtered EEG. Finally, a linear support vector machine (SVM) is trained by the extracted features. This approach is called class discrepancy-guided sub-band filter-based CSP (CDFCSP) because the CDF is utilized to mend the CSP algorithm in this paper. The experimental results for the BCI competition IV dataset 2a and dataset 2b show that the CDF can improve the performance of the CSP algorithm and the CDFCSP algorithm outperforms other compared state-of-the-art algorithms.

This paper is organized as follows. Section 1 presents the introduction. Section 2 describes the methodology of the proposed CDFCSP and the datasets and the experiment. Section 3 presents the experimental results and discussion. Section 4 provides the discussion and conclusions.

## 2. Materials and methods

The block diagram of the proposed CDFCSP-based approach is presented in Fig. 1.

First, the energy spectrum of the training EEG signal is extracted by WPD, and the energy spectrum templates are obtained by averaging the energy spectra of the trials in each class. The energy spectrum templates are exploited to guide the design of the CDF. Second, the segmented training EEG signal is filtered by the CDF to augment the discriminative frequency bands with large energy differences between class templates, and then the FBCSP algorithm (Novi et al., 2007) is used to extract the features. Third, a linear SVM classifier is trained using the extracted feature vectors. Finally, the features of the tested EEG signals extracted via segmentation, the CDF, and the FBCSP are fed to the trained SVM to obtain a prediction. The details of the algorithm are presented as follows.

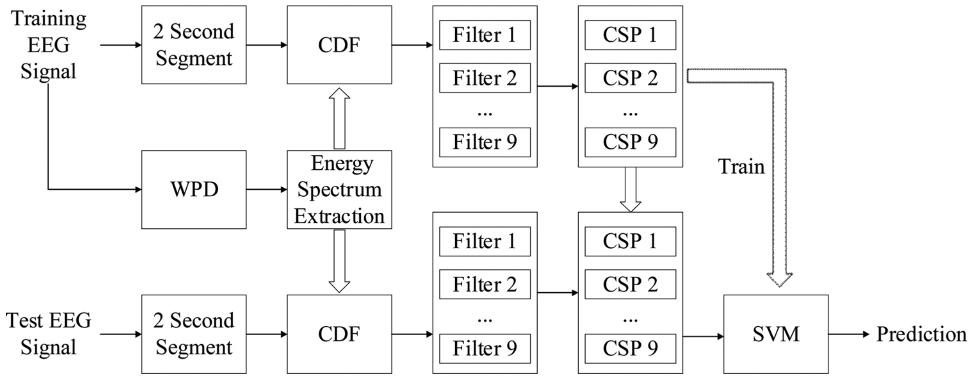



**Fig. 1.** Block diagram of the proposed CDFCSP method.

### 2.1. Energy spectrum template

To obtain the designing guideline of the CDF, the energy spectrum template was calculated. The energy distribution in the time domain and frequency domain can be given simultaneously by a time-frequency analysis method. WPD is a popular time-frequency analysis method and offers an accurate time-frequency representation, a good noise reduction performance, and a good trade-off between time and frequency resolution (Daubechies, 1990, 1992). Therefore, WPD has been widely applied in transient and non-stationary signal processing (Feng et al., 2015), such as for EEG signals (Qin and He, 2005; Sun et al., 2017). In this paper, WPD was applied as the time-frequency analysis method. However, the proposed CDF can be easily generalized to the cases of other time-frequency analysis tools.

To obtain the content-aware bandpass filter, the energy spectrum templates were calculated by averaging the energy spectra of all trials in each class. Let $WP^m(i, k)$ denote the WPD coefficients in the $j$th level from the $m$th channel and $i = 1, \cdots, I$ and $k = 1, \cdots K$ denote the index of the frequency components and the index of coefficients in the frequency components, respectively. The square of the coefficients in each frequency component were summed, which represented the energy spectrum of the signal from the $m$th channel:

$$s(m, i) = \sum_{k=1}^{K} (WP^m(i, k))^2 \tag{1}$$

For multiple channel EEGs, the energy spectrum template in the EEG channel $m$ and the frequency component $i$ of class $c$ (the training set of class c is C) was calculated using an averaging method among the trials in class $c$:

$$T_c(m, i) = \frac{1}{N} \sum_{trials \in C} s(m, i) \tag{2}$$

where $N$ is the number of trials in the training set of class c. The energy spectrum templates of all channels of the EEG were calculated to guide the CDF. The energy spectrum template of subject 3 in the BCI competition IV dataset 2a is shown in Fig. 2. Three EEG channels (C3, Cz and C4 in the international 10/20 system) are included in the figure, and additional details about the dataset can be found in Section 2.5.

Fig. 2 shows that (1) the energy of certain frequency components (such as frequency components 3 and 4 in channels C3 and C4, indicating 8 Hz–16 Hz) is different between the left- and right-hand motor imagery and thus is more discriminative for classification; (2) the energy of certain frequency components (such as the frequency components in channel Cz) is nearly equivalent between the left- and right-hand motor imagery and thus is less discriminative for classification.

### 2.2. Class discrepancy-guided sub-band filter for pre-processing

The class discrepancy-guided sub-band filter (CDF) is proposed for preprocessing the EEG in this section. Through the analysis above, the frequency components with significant class templates differences should be more discriminative. Ideally, the proposed filter should augment the discriminative frequency components and suppress the less discriminative frequency components automatically before feature extraction. Thus, the filter should improve the performance of the CSP algorithm. Therefore, the energy spectrum template obtained acts as the designing guideline of the CDF.

The CDF based filtering is performed by changing the WPD coefficients in specific frequency components. Initially, the WPD of the EEG segment was applied to obtain the WPD coefficients. Let $WP_{in}^m(i, k)$ be the WPD coefficients of the original EEG segment in channel $m$ and frequency component $i$. Then, the WPD coefficients were enlarged or shrank based on the CDF matrix $H(m, i)$ as follows:

$$WP_{out}^m(i, k) = H(m, i) \cdot WP_{in}^m(i, k) \tag{3}$$

$H(m, i)$ is the CDF matrix and $WP_{out}^m(i, k)$ is the WPD coefficients of the filtered signal. Finally, the modified WPD coefficients $WP_{out}^m(i, k)$ were used to reconstruct the EEG segment through the inverse WPD. Thus, the output EEG filtered by CDF is achieved. Specifically, the CDF matrix is formulated in two modes: ratio mode and normalization mode.

#### 2.2.1. Ratio mode

Because the CDF is based on the effectiveness of frequency components with different energies between classes for classification, the sub-band filters in channel $m$ and component $i$ were designed as follows:

$$H_1^R(m, i) = \frac{T_{class1}(m, i)}{T_{class2}(m, i)} \tag{4a}$$

$$H_2^R(m, i) = \frac{T_{class2}(m, i)}{T_{class1}(m, i)} \tag{4b}$$

Therefore, the value in the filter vector is a ratio of one class's template to the other. With this filter, the frequency bands of the input signal are augmented or depressed according to the a priori knowledge of the samples in each class, i.e., the template difference between the classes. Both sub-band filters ((4a) and (4b)) were applied to filter the raw EEG signal. If the energy spectrum templates of a frequency band are significantly different between classes, then this frequency band should be discriminative. Therefore, one of the sub-band filters would be large, which could enlarge the amplitude of the coefficients in related frequency components and augment this frequency component. If the energy spectrum templates are nearly equivalent between two classes, even if the energy is high, the frequency band should be less discriminative. In this situation, two sub-band filters will both be close to 1 and no components will be augmented. As showed in Fig. 2, for subject 3 in the BCI competition IV dataset 2a, the frequency components 3 and 4 in channel C3 and C4 will be augmented through the CDF.

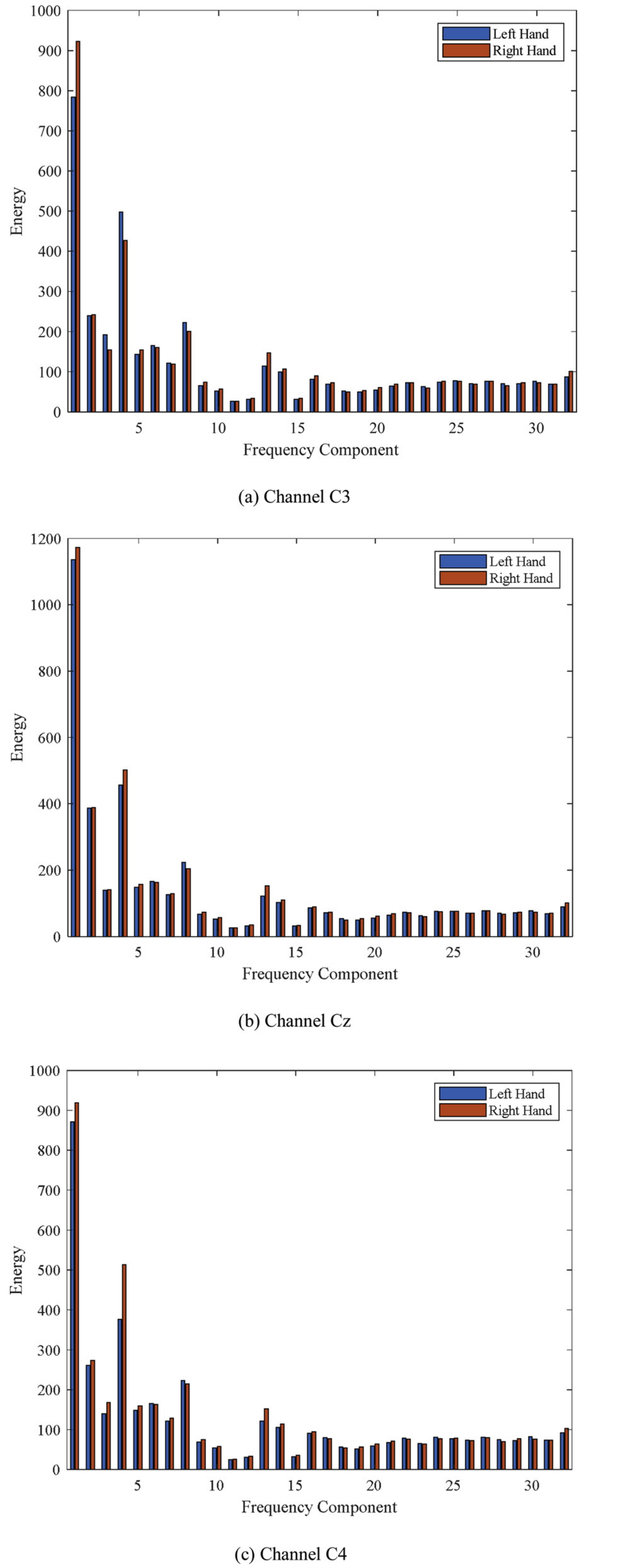



**Fig. 2.** Energy spectrum template of subject 3 in the BCI competition IV dataset 2a.

After the filtering process, the filtered EEG signal was reconstructed via inverse WPD based on $WP_{out}(i, k)$. Because two sub-band filters were employed, two filtered EEG signals were obtained from one EEG channel. Thus, the CDF doubled the channel number of the EEG so that more channel data would be available to be imposed on the CSP algorithm.

### 2.2.2. Normalization mode

Another mode forming the CDF is the normalization mode:

$$H_1^N(m, i) = \frac{T_{class1}(m, i)}{T_{class1}(m, i) + T_{class2}(m, i)} \tag{5a}$$

$$H_2^N(m, i) = \frac{T_{class2}(m, i)}{T_{class1}(m, i) + T_{class2}(m, i)} \tag{5b}$$

According to (5a) and (5b), the value in the filter vector is the normalized template. Similar to the ratio mode, both of the sub-band filters (5a and 5b) were employed to filter the raw EEG signal and the energy in frequency bands was augmented when the templates were significantly different between classes.

According to (4a), (4b), (5a) and (5b), the CDF is determined by the difference between the class templates, i.e., the averaged energy spectrum of the sample in each class. Certain CDF properties have been identified.

(1) The CDF is a class discrepancy-guided sub-band filter that aims to recognize and augment the discriminative frequency bands. This filter should improve the performance of the CSP algorithm because the frequency bands with distinctive templates between classes are more discriminative, and these characteristics hold true for most subjects.
(2) The filter is designed for each subject based on the a priori knowledge extracted from the training set of the subject.
(3) The channel number of the EEG will be doubled because two filters in the CDF are applied. More channel data will obviously correct the performance of the CSP algorithm in a few-channel situation.

## 2.3. Filter bank common spatial pattern for feature extraction

CSP based feature has been widely applied in two-class motor imagery classification (Zhang et al., 2017). CSP algorithm searches for spatial filters which maximize the variance of projected signal from one class and meanwhile minimize the variance of projected signal from the other class. CSP based feature from multiple frequency bands presents the ERD/ERS phenomena in multiple frequency bands and has been proven effective in motor imagery-based EEG classification (Ang et al., 2012). Therefore, the FBCSP was used to extract features. Specifically, the CSP features were extracted based on the EEG filtered by filter bank.

Several sub-band bandpass filters decomposed the EEG into multiple-filtered EEGs before the CSP algorithm was applied. Causal Chebyshev Type II bandpass filters, including 9 sub-bands covering 4–40 Hz (4–8 Hz, 8–12 Hz, 12–16 Hz, …, 36–40 Hz), were chosen in this paper (Ang et al., 2012). Narrow bandpass frequency ranges will increase the filter order and the computing complexity, whereas a wider frequency range will lose resolution (less number of frequency bands and less features). Therefore, the frequency range of the sub-band bandpass filter represents a compromise.

The CSP algorithm extracts features from the filtered EEG. EEG trials are represented as the matrix $E$ of size $M \times L$, where $M$ is the number of channels in the EEG and $L$ is the number of sampling points in a trial. Initially, the normalized covariance matrixes of the EEGs of the two classes were calculated as follows:

$$C_1 = \frac{E_1 E_1^T}{trace(E_1 E_1^T)}$$
$$C_2 = \frac{E_2 E_2^T}{trace(E_2 E_2^T)} \tag{6}$$

Then, the average covariance matrixes of the two classes were used to calculate the composite spatial covariance matrix as follows:

$$R = \overline{C_1} + \overline{C_2} \tag{7}$$

The composite spatial covariance matrix $R$ is an $M$-order square matrix and could be decomposed using eigenvalues and eigenvectors as follows:

$$R = V_0 D_0 V_0^T \tag{8}$$

where $D_0$ is the diagonal matrix of eigenvalues and $V_0$ is the matrix of related eigenvectors. Then, the whitening transformation matrix was calculated using eigenvalues and eigenvectors:

$$W = \mathrm{D}_0^{-\frac{1}{2}} \mathrm{V}_0^T \tag{9}$$

The whitening transformation was conducted to whiten the spatial covariance:

$$S_1 = W\overline{C_1}W^T$$
$$S_2 = W\overline{C_2}W^T \tag{10}$$

The whitened spatial covariance matrix was decomposed using eigenvalues and eigenvectors as follows:

$$S_1 = VD_1V^T$$
$$S_2 = VD_2V^T$$
$$D_1 + D_2 = I \tag{11}$$

where the eigenvectors $V$ of two classes are same. The sum of the corresponding eigenvalues $D_1$ and $D_2$ is equal to 1. The eigenvectors related to the $e$ largest eigenvalues in $D_1$ and $D_2$ form the projection matrix:

$$P = (V')^T W \tag{12}$$

where V' is constituted by the eigenvectors related to the largest $e$ eigenvalues and smallest $e$ eigenvalues in $D_1$. Therefore, V' is a $M \times 2e$ vector. The EEG was transformed using the projection matrix into a $2e \times M$ vector $Z_t$:

$$Z_t = P \times E_t \tag{13}$$

The logarithm of the normalized variance in EEG trial $Z_t$ was then calculated as the features to be classified:

$$f_{t,j} = \log\left(\frac{\mathrm{var}(Z_{t,j})}{\sum_{j=1}^{2e} \mathrm{var}(Z_{t,j})}\right),\ t = 1, 2;\ j = 1, 2, \cdots, 2e \tag{14}$$

Because *2e* features could be extracted from a single filtered EEG and a filter bank including 9 bandpass filters was used, *18e* features were obtained from each trial by the FBCSP algorithm.

### 2.4. Support vector machine classifier

A number of classifiers have been employed by researchers for different types of BCI systems, including linear discriminant analysis (LDA) classifiers, Bayesian-based classifiers, SVMs, neural networks and combinations of classifiers. Reviews have described the comparisons between classifiers leveraged in BCI fields (Lotte et al., 2007; Szczuko et al., 2018). SVM is simple, robust, efficient and widely applied in EEG analysis; therefore, it is recommended by researchers in the field of BCI based on EEG analysis (Lotte et al., 2007; Quitadamo et al., 2017). Because the principle of the CSP is to maximum the variance in one class and minimum the variance in the other simultaneously, features are extracted based on the normalized variance. Therefore, the CSP features are expected to be linearly separable, and a linear SVM was chosen in this paper. The SVM classifier in MATLAB R2018a was employed in the following experiment. In addition, “boxConstraint” in MATLAB is the maximum penalty imposed on margin-violating observations, and it is usually called a penalty multiplier. “KernelScale” specifies the value by which all elements of the predictor matrix are divided. These two hyperparameters influence the performance significantly; therefore, the minimization of the cross-validation loss was applied to find the optimal values as described in Section 3.4.

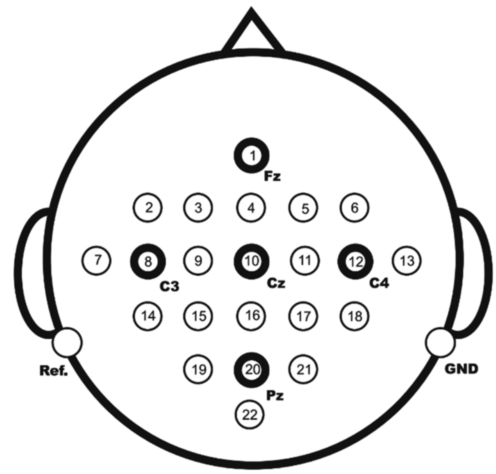



**Fig. 3.** Electrode montage corresponding to the international 10–20 system used in the BCI competition IV dataset 2a.

### 2.5. Dataset description

The datasets applied to validate the efficiency of the proposed CDFCSP included the competition IV dataset 2a and dataset 2b.

#### 2.5.1. BCI competition IV dataset 2a

Motor imagery-based EEGs from 9 subjects were included in this dataset (Tangermann et al., 2012; Brunner et al., 2008), including one session with 288 trials for training and another session with 288 trials for testing. Four-category motor imagery EEGs (left hand, right hand, tongue and both feet) from 22 channels (international 10–20 system, as shown in Fig. 3) were recorded. To evaluate the CDFCSP performance in a few-channel situation, only the EEGs from channels C3, C4 and Cz were used in the following experiment to accomplish a two-category classification, i.e., left-hand and right-hand motor imagery. The data were sampled at 250 Hz, and the sensitivity of the amplifier was 100 μV. The length of the trial was 7.5 s, and 4 s of the EEG were motor imagery related. A 0.5 Hz to 100 Hz bandpass filter and a 50 Hz notch filter were imposed to filter the EEGs, and the trials with artefacts were marked by experts. Predictions were performed for each sampling point, and the classification performance was evaluated by the maximum kappa value over the time course. The kappa value is a more robust evaluation criterion that considers chance agreement (A Schog, 2007). Because the sample number in each class was equal, the kappa value was transferred from the classification accuracy as follows:

$$\mathrm{kappa} = \frac{C \times P - 1}{C - 1} \tag{15}$$

where $C$ is the category number in the classification and $P$ is the accuracy.

#### 2.5.2. BCI competition IV dataset 2b

This dataset includes EEGs for two categories of motor imagery (left hand and right hand) from 9 subjects (Tangermann et al., 2012; Leeb et al., 2008). Five sessions were provided per subject, with the first 3 sessions used as training sets and the last 2 sessions used as test sets. The number of trials in each session was between 120 and 160. Only trials in sessions 3, 4 and 5 contained feedback procedures; therefore, session 3 was used as the training set and session 4 and session 5 were applied to evaluated the experimental results as implemented in a previous study (Shahid and Prasad, 2011). Channels C3, Cz and C4 (international 10–20 system) were employed to collect the EEG at 250 Hz, and the dynamic range of the feedback sessions was $\pm\ 50\,\mu V$. Similar to dataset 2a, the EEGs were filtered by a 0.5 Hz to 100 Hz bandpass filter and a 50 Hz notch filter and the artefacts were marked by experts. Additionally, the performance of the classification was evaluated by the maximum kappa value.

### 2.6. Experiment description

Because the classification of motor imagery-based EEGs is a small-sample problem and the number of trials in the training sets was limited, the trials in the training set were segmented into 2 s segments, with 1.9 s overlapping to enlarge the training set and prevent overfitting. One segment was treated as a sample in the following experiment. To obtain the classification output at every sampling point and build a causal system, the 2-second EEG before a sampling point was used to generate predictions. Because the adjacent signal is nearly equivalent, predictions were performed every 10 sampling points to reduce the computing load as same as the FBCSP algorithm (Ang et al., 2012).

In the WPD procedure, the level of decomposition will determine the resolution of the energy spectrum. The decomposition level is a key parameter, and related experiments were conducted in Section 3 to search for a good choice. FBCSP-based approaches (Ang et al., 2012) leverage a 4 Hz bandpass filter to preprocess the EEG, and the resolution was similar to that of a 5-level WPD because a 5-level WPD will obtain $2^5$ components and the width of the frequency band was approximately 4 Hz. Therefore, decomposition levels of approximately 5 (4–7) were tested. In addition, the wavelet basis was set to 8 dB according to the experiment comparison in (Luo et al., 2016).

## 3. Results

### 3.1. Discriminative frequency components augmented by the CDF

In the analysis presented in Section 2, the CDF exploits the a priori knowledge of the samples in each class and the difference between classes' templates were used as the guidelines for the sub-band filter. The CDF bandpass coefficient $H(m, i)$ in each channel and frequency band (presented as a rectangle) for subjects in the BCI competition dataset 2a were calculated and are presented in Fig. 4. Because the value obtained from (4a) and (4b) were reciprocals of each other, only the larger one is shown in the figure. The colour is brighter when the value is larger as indicated by the legend. For example, the frequency band at 20–24 Hz from channel C4 was the most discriminative for subject A01 and the bandpass coefficient was approximately 1.16, whereas the frequency band at 12–16 Hz from channel C4 was the most discriminative for subject A03 and the bandpass coefficient was approximately 1.32. The discriminative frequency bands detected by the CDF for each subject were obviously different. Thus, the CDF recognizes frequency bands with large inter-class differences. Because the value obtained from (4a) and (4b) were applied as the bandpass coefficients, the frequency bands with large inter-class differences were augmented when the EEG was filtered by the CDF. Therefore, we can conclude that the CDF is a content-aware bandpass filter.

### 3.2. Performance comparison of the FBCSP with and without the CDF

To validate the assumption that features extracted from the EEG filtered by the CDF would perform better in classification, experiments were performed on the BCI competition IV dataset 2a and dataset 2b. The classification procedures were applied as described in Section 2, and the use of the CDF was the only difference in the experimental setting. Table 1 presents a comparison of the experimental results in which the maximum kappa value was employed as the evaluating criterion. As mentioned above, predictions were performed every 10 sampling points. 'FBCSP' represents the algorithm without CDF, whereas 'CDFCSP(R)' and 'CDFCSP(N)' represent the ratio mode CDFCSP and normalization mode CDFCSP, respectively. The subjects in dataset 2a were indicated as A01 to A09, and the subjects in dataset 2b were indicated as B01 to B09. The highest kappa value is marked in bold.

Student's *t*-test was also conducted to confirm the significance of the performance improvement, and the results are presented in Table 2. The *p*-values of the ratio mode CDFCSP VS. FBCSP and normalization mode CDFCSP VS. FBCSP were 0.040 and 0.032, respectively, and these results illustrate that the performance improvement provided by the CDF was significant.

Several conclusions can be drawn from the experimental results given in Tables 1 and 2: (1) the CDF can strengthen the classification ability of FBCSP-based features significantly; (2) the CDFCSP-based approach perform best (or was at least comparable) on most subjects; and (3) the experimental results of the ratio mode CDFCSP and normalization mode CDFCSP were similar. The experimental results mentioned above verify the performance improvement of the CDF.

### 3.3. Performance comparison of the CDFCSP and state-of-the-art algorithms

The performance of state-of-the-art algorithms was also compared with the proposed CDFCSP method. The classification results of the recurrent quantum neural network (RQNN)-based approach (Gandhi et al., 2014), the dynamic frequency feature selection (DFFS)-based approach (Luo et al., 2016), DCSP-based approach (Thomas et al., 2009), IDFBCSP-based approach (Kumar et al., 2017b), TFPOCSP-based approach (Kumar and Sharma, 2018), WPDCSP-based approach (Yang et al., 2016), SBL-based approach (Zhang et al., 2017), SFTO-based approach (Miao et al., 2017), CSP-TSM based approach (Kumar et al., 2017a) and CDFCSP (ratio mode denoted as CDFCSP(R) and normalization mode denoted as CDFCSP(N))-based approaches on the BCI competition IV dataset 2a and BCI competition IV dataset 2b are presented in Table 3. Because the CDF is proposed to improve the performance of CSP in few-channel EEG situation, only the EEGs from channels C3, Cz and C4 were applied to achieve a left-hand and right-hand two-category motor imagery classification in dataset 2a. Because the SFTO based approach utilized 4.5 s EEG trial in training and evaluating stage, only one kappa value was obtained over the time course.

The comparison in Table 3 indicates that (1) the proposed CDFCSP-based approaches obtain the best average performance; (2) the two modes of CDFCSP are superior to the compared algorithms and present an average increase of 20%, 15%, 25%, 7%, 15%, 28%, 7%, 20%, and 13% in kappa value; (3) the ratio mode and normalization mode CDFCSP behave similarly.

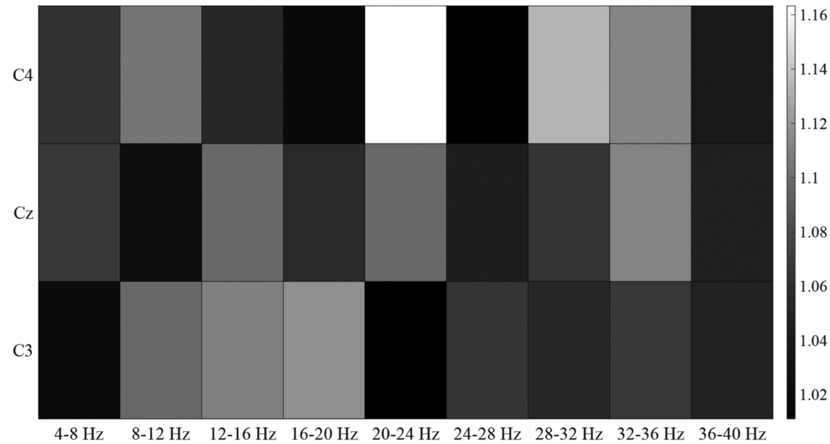
### 3.4. Parameter selection

The level of WPD that determines the resolution of the CDF was a key parameter. The resolution of the filter can be calculated as follows:

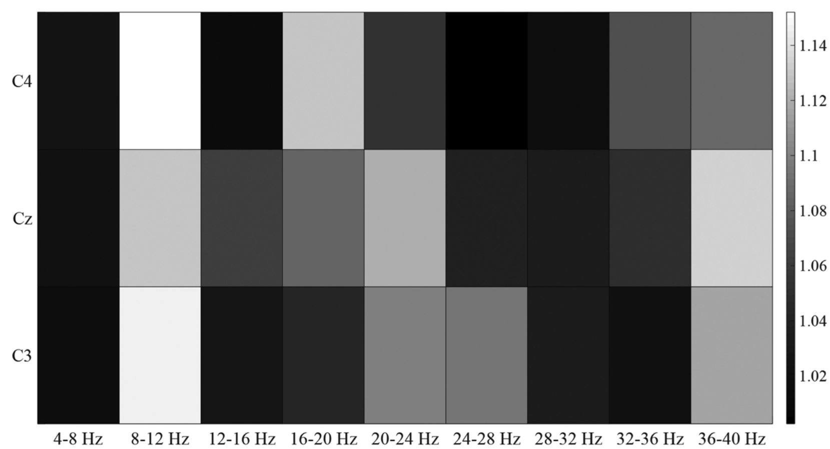
$$Re = \frac{Fs}{2 \times Fc} \tag{16}$$

where $Fs$ is the sampling frequency and $Fc$ is the number of frequency components. For example, if a 5-level WPD can obtain 32 frequency components and the sampling frequency of the EEG is 250 Hz, then the resolution of the filter will be approximately 4 Hz. Higher level WPDs will obtain more frequency components and a higher resolution filter, but the computing cost is more expensive. One additional level in the WPD will double the frequency component number and the computing complexity of the CDF. The classification results of the CDFCSP applying different WPD levels are shown in Table 4. Because of the similar capabilities between the ratio mode and normalization mode CDFCSP, only the ratio mode approach was included in the comparison. The results show that level 5 obtains the highest kappa value; thus, a 5 level WPD was applied in this paper.

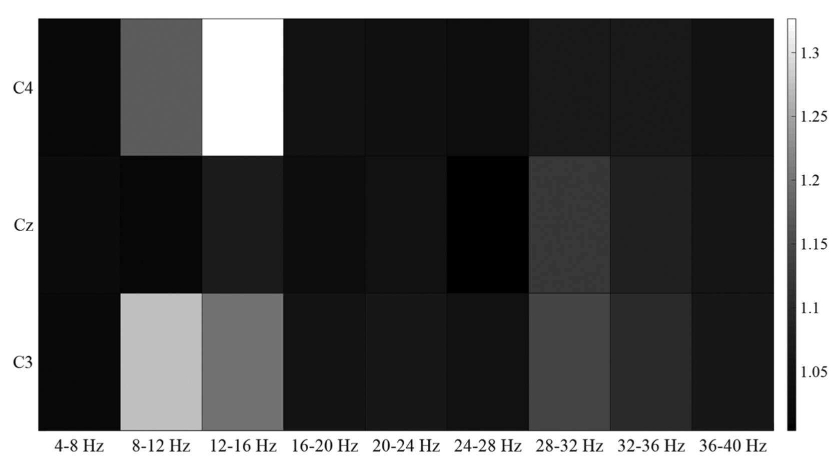
As mentioned in Section 2.4, "boxConstraint" and "KernelScale", which are hyperparameters in MATLAB's SVM classifier, greatly




(a) Subject A01




(b) Subject A02




(c) Subject A03

**Fig. 4.** Class template differences in the subjects from the BCI competition IV dataset 2a. The horizontal axis indicates the frequency bands, and the vertical axis indicates the EEG channels. The brightness of each area describes the bandpass coefficients as indicated by the legend.

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**Table 1**
Comparison of the performance (max kappa value) using the FBCSP with and without the CDF.

| Subject | FBCSP | CDFCSP(R) | CDFCSP(N) |
|---|---|---|---|
| A01 | 0.6028 | 0.6028 | **0.6170** |
| A02 | **0.1972** | **0.1972** | **0.1972** |
| A03 | 0.8248 | **0.8686** | **0.8686** |
| A04 | 0.3793 | **0.3966** | **0.3966** |
| A05 | 0.3778 | **0.4667** | **0.4667** |
| A06 | 0.3519 | **0.3704** | **0.3704** |
| A07 | 0.4714 | **0.5000** | **0.5000** |
| A08 | **0.8657** | 0.7612 | 0.7761 |
| A09 | 0.6154 | **0.6615** | **0.6615** |
| B01 | 0.3864 | 0.4566 | **0.4831** |
| B02 | 0.2246 | **0.2834** | **0.2876** |
| B03 | **0.3227** | 0.2608 | 0.2516 |
| B04 | 0.8043 | **0.9675** | **0.9675** |
| B05 | 0.9038 | **0.9850** | 0.9765 |
| B06 | **0.7488** | 0.6991 | 0.7005 |
| B07 | 0.6638 | **0.6691** | **0.6691** |
| B08 | **0.8646** | 0.8454 | 0.8375 |
| B09 | 0.6234 | **0.8084** | 0.7920 |
| Average | 0.5683 | 0.6000 | **0.6011** |

**Table 2**
Evaluation of the statistical significance of the performance difference based on Student's *t*-test.

| | CDFCSP(R) VS. FBCSP | CDFCSP(N) VS. FBCSP |
|---|---|---|
| *p*-Values | 0.040 | 0.032 |

influenced the classification accuracy. Our results showed that the hyperparameters minimize the five-fold cross-validation losses by using Bayesian optimization; moreover, the cross-validation loss is equivalent to the classification loss for observations not used for training. For the sake of reproducibility, the selected parameters for subjects in the BCI competition IV dataset 2a are listed in Table 5. The best parameters vary from subject to subject. In general, small "boxConstraint" and "KernelScale" values tend to produce good performances except with subject A02, whose "boxConstraint" value was 986.25.

In addition, Fig. 5 shows the best objective function value against the iteration number of optimization for subject A01. In Fig. 5, the $x$-axis represents the iteration number of optimization and the $y$-axis represents the objective function value. The objective function value decreased as the iteration progressed, and the minimum objective function value appeared at the 16th iteration. After this point, the objective function did not decrease further.

**Table 4**
Classification results (max kappa value) of the ratio mode CDFCSP applying different WPD levels.

| Basis | Level 3 | Level 4 | Level 5 | Level 6 | Level 7 |
|---|---|---|---|---|---|
| accuracy | 0.53 | 0.53 | **0.54** | 0.51 | 0.49 |

**Table 5**
Selected parameters for subjects in the BCI competition IV dataset 2a.

| Subject | BoxConstraint | KernelScale |
|---|---|---|
| A01 | 29.35 | 1.5984 |
| A02 | 986.25 | 2.918 |
| A03 | 0.0018272 | 0.0046904 |
| A04 | 0.22256 | 0.067974 |
| A05 | 0.0020306 | 0.022971 |
| A06 | 3.5955 | 2.1401 |
| A07 | 0.58588 | 0.077901 |
| A08 | 0.0010001 | 0.019686 |
| A09 | 1.9099 | 0.25438 |

Fig. 6 presents a model of the objective function against the parameters for subject A01. The $x$-axis represents the parameter "boxConstraint", the $y$-axis represents the parameter "KernelScale" and the $y$-axis represents the objective function value. Blue points are the observed points, and the green star indicates parameters with the least objective function. We can conclude from Fig. 6 that the objective function graph is a slope and a large "boxConstraint" value and small "KernelScale" value obtains high objective function.

## 4. Discussions

This paper proposes CDF and CDFCSP methods to improve the CSP algorithm applied in motor imagery-based EEG classification. The CDF can automatically and efficiently recognize and augment the discriminative frequency bands and reasonably extends the channel numbers of EEGs. Additionally, the CDF takes advantage of the a priori knowledge of each class, and the differences in the class templates are used as the guideline for sub-band filtering. Because frequency bands with large energy differences between classes are more distinguishable, the CDF recognizes and augments the frequency bands according to WPD energy templates. The FBCSP algorithm was used to extract

**Table 3**
Comparison of the performance (max kappa value) of the proposed CDFCSP-based approach with other state-of-the-art algorithms.

| Subject | RQNN | DFFS | DCSP | IDFB CSP | TFPO CSP | WPD CSP | SBL | SFTO | CSP- TSM | CDFCSP (R) | CDFCSP (N) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| A01 | 0.22 | 0.27 | 0.46 | 0.56 | 0.67 | 0.56 | 0.59 | 0.50 | 0.60 | 0.60 | 0.62 |
| A02 | 0.22 | 0.24 | 0.24 | 0.21 | 0.18 | 0.23 | 0.01 | 0.17 | 0.17 | 0.20 | 0.20 |
| A03 | 0.58 | 0.82 | 0.20 | 0.87 | 0.77 | 0.77 | 0.85 | 0.72 | 0.81 | 0.87 | 0.87 |
| A04 | 0.21 | 0.23 | 0.09 | 0.34 | 0.43 | 0.24 | 0.36 | 0.17 | 0.45 | 0.40 | 0.40 |
| A05 | 0.43 | 0.27 | 0.17 | 0.21 | 0.16 | 0.14 | 0.44 | 0.44 | 0.13 | 0.47 | 0.47 |
| A06 | 0.22 | 0.32 | 0.31 | 0.35 | 0.39 | 0.34 | 0.41 | 0.33 | 0.33 | 0.37 | 0.37 |
| A07 | 0.17 | 0.19 | 0.51 | 0.46 | 0.27 | 0.20 | 0.49 | 0.33 | 0.37 | 0.50 | 0.50 |
| A08 | 0.35 | 0.26 | 0.60 | 0.87 | 0.82 | 0.70 | 0.81 | 0.75 | 0.85 | 0.76 | 0.78 |
| A09 | 0.58 | 0.69 | 0.74 | 0.82 | 0.83 | 0.59 | 0.63 | 0.71 | 0.69 | 0.66 | 0.66 |
| B01 | 0.55 | 0.46 | 0.31 | 0.30 | 0.33 | 0.31 | 0.41 | 0.44 | 0.36 | 0.46 | 0.48 |
| B02 | 0.27 | 0.35 | 0.20 | 0.21 | 0.00 | 0.29 | 0.13 | 0.17 | 0.33 | 0.28 | 0.29 |
| B03 | 0.72 | 0.26 | 0.30 | 0.25 | −0.06 | 0.29 | 0.28 | 0.06 | 0.16 | 0.26 | 0.25 |
| B04 | 0.96 | 0.95 | 0.90 | 0.90 | 0.95 | 0.92 | 0.87 | 0.82 | 0.93 | 0.97 | 0.97 |
| B05 | 0.83 | 0.91 | 0.83 | 0.82 | 0.68 | 0.70 | 0.89 | 0.73 | 0.54 | 0.99 | 0.98 |
| B06 | 0.71 | 0.73 | 0.60 | 0.66 | 0.76 | 0.36 | 0.72 | 0.53 | 0.65 | 0.70 | 0.70 |
| B07 | 0.46 | 0.69 | 0.67 | 0.67 | 0.62 | 0.56 | 0.60 | 0.61 | 0.58 | 0.67 | 0.67 |
| B08 | 0.84 | 0.92 | 0.86 | 0.88 | 0.77 | 0.79 | 0.89 | 0.73 | 0.86 | 0.85 | 0.84 |
| B09 | 0.75 | 0.85 | 0.67 | 0.73 | 0.75 | 0.50 | 0.68 | 0.71 | 0.72 | 0.81 | 0.79 |
| Ave | 0.50 | 0.52 | 0.48 | 0.56 | 0.52 | 0.47 | 0.56 | 0.50 | 0.53 | **0.60** | **0.60** |

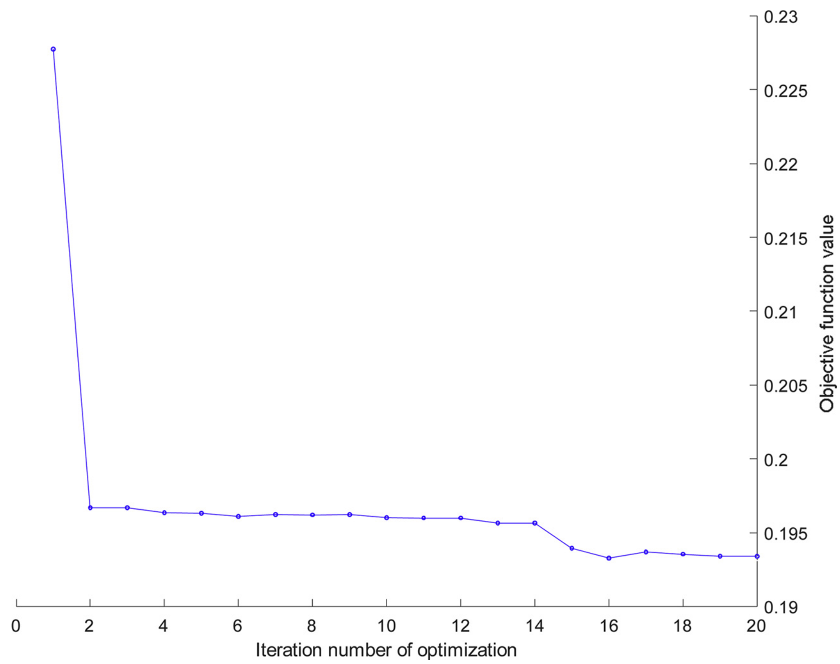


**Fig. 5.** Best objective function value against the iteration number of optimization for subject A01.

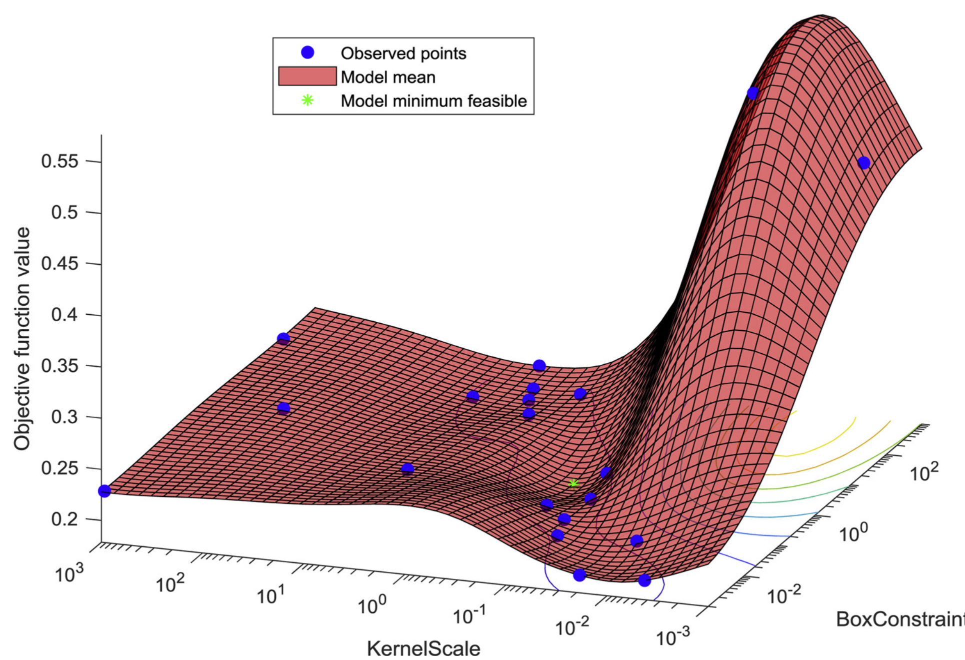


**Fig. 6.** Model of the objective function against the parameters.

features from the CDF-filtered EEGs, and then the features were fed into a linear SVM classifier. The modified algorithm was called the CDFCSP, which obtained promising classification results in motor imagery-based EEG classification. The experimental results given in Section 3 show the efficiency of the proposed CDF and CDFCSP methods.

However, when multiple channel EEG samples were applied in an experiment, the performance improvement provided by the CDF was slight, and this result was associated with the main drawback of CDFCSP, which limits the application range.

The proposed algorithm is designed for a synchronous BCI that includes indications of the onset time of motor imagery. To generalize the use of the CDFCSP in asynchronous BCIs (real-time BCIs), a timing detection algorithm must be added to the system (Gandhi et al., 2014). This motor imagery detection approach represents our next stage of work.

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